

Muhammad Anas Akhtar

AI Engineer | Generative AI & MLOps Specialist

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PROFESSIONAL SUMMARY

Innovative **AI & MLOps Engineer** specializing in **Generative AI** and **Agentic Workflows**. Expert in architecting **RAG pipelines** and multi-agent systems using **LangGraph** and **CrewAI** on **AWS/GCP**. Proven track record in deploying production-grade solutions with robust observability via **LangSmith**.

TECHNICAL SKILLS

Generative AI & LLMs — Expert

Large Language Models (LLMs), LLaMA-2, OpenAI, RAG Architecture, Fine-tuning (PEFT/LoRA), Prompt Engineering, NLP

Agentic Frameworks: — Expert

LangGraph, **LangSmith**, **LangChain**, CrewAI, DSPy, LlamaIndex, PromptFlow, Chain-of-Thought (CoT)

MLOps & Cloud Infrastructure: — Expert

MLOps, **AWS SageMaker**, GCP Vertex AI, Docker, CI/CD, Model Monitoring, FastAPI

Data Engineering & Tools: — Expert

Python (Expert), SQL, **Data Engineering**, n8n, Make.com, Pinecone, ChromaDB, Weaviate.

PROFESSIONAL EXPERIENCE

Generative AI Engineer

01/2024 – Present

- **Architected** scalable **RAG solutions**, improving retrieval accuracy by **15%** via optimized vector search.
- **Developed** autonomous workflows using **LangGraph** and managed evaluation via **LangSmith**.
- **Deployed** AI models on **AWS/GCP** using **MLOps** for automated scaling and monitoring.
- **Fine-tuned** LLaMA-2/OpenAI models using **PEFT/LoRA** for enterprise automation solutions.

Machine Learning Engineer

02/2022 – 12/2023

- **Engineered** end-to-end ML pipelines for **NLP** and **Predictive Analytics** in global startup environments.
- **Optimized** model inference through hyperparameter tuning and compression, reducing real-time latency.
- **Containerized** ML workflows with **Docker**, ensuring seamless CI/CD integration across cloud infrastructures.

PROJECTS

AI-Powered Medical Document QA Chatbot (RAG-Based System) 🔗

AI Voice Cloning & Text-to-Speech System 🔗

AI-Based Smart Interview Assistant for Candidate Evaluation & Screening 🔗

EDUCATION

Bachelor of Science in Artificial Intelligence

2020 – 2024

The Islamia University of Bahawalpur, Pakistan

CERTIFICATES

• Generative AI & Transformers – Coursera

• LangChain for LLMs – Coursera

• Foundational Generative AI – iNeuron

• Enterprise Generative AI – IBM

• AWS & Google Cloud for ML – Udemy